



# Ninaivu

நினைவு · memory

**For whoever looks after the technical side.**

Technical administrator guide · version 1.0.5

**Docker**

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# Technical administrator guide

The guide tells a household how to install Ninaivu with the installer for their computer and use it. This page is for whoever looks after the technical side: the source code, Docker, running from a checkout, the command line, environment variables and where to report a problem. It is in English only.

## Where things are

|                                                    |                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Source code                                        | <a href="https://github.com/javajaga-usa/Ninaivu">https://github.com/javajaga-usa/Ninaivu</a>                                           |
| Installers (the download page the guide points to) | <a href="https://github.com/javajaga-usa/Ninaivu/releases/latest">https://github.com/javajaga-usa/Ninaivu/releases/latest</a>           |
| Reporting a problem                                | <a href="https://github.com/javajaga-usa/Ninaivu/issues">https://github.com/javajaga-usa/Ninaivu/issues</a>                             |
| Extensions (Creative Studio, Gemini and the rest)  | <a href="https://github.com/javajaga-usa/Ninaivu/tree/main/extensions">https://github.com/javajaga-usa/Ninaivu/tree/main/extensions</a> |
| The docs site                                      | <a href="https://javajaga-usa.github.io/Ninaivu/">https://javajaga-usa.github.io/Ninaivu/</a>                                           |

When reporting a problem, say which version (**System → Home & extensions** shows it under *This home*), what was done, and what the log said.

## More on the installers

- **Windows.** Once Ninaivu is in the winget catalogue, `winget install Ninaivu.Ninaivu` will work too.
- **macOS.** Once Ninaivu is in Homebrew, `brew install --cask ninaivu` will work too. `open -a Ninaivu --args --tray` opens the menu-bar tray instead of the Control Panel.
- **Linux and Raspberry Pi.** `--photos DIR`, `--prefix DIR`, `--no-service` and `--quiet` skip the installer's questions, for a headless Pi set up over SSH. Running it again upgrades: the photographs folder the service was given is kept (pass `--photos` to change it), and so are the index, the settings and the AI models (`~/.local/state/ninaivu/ai-models`). The service is told its state folder (`NINAIVU_STATE_DIR`); an install from 1.0 whose data is in `~/.ninaivu` keeps using it there. `~/.local/share/ninaivu/uninstall` removes the program, the commands, the menu entry and the service, and only those: a `--prefix` folder with other things in it is left with them. `uninstall --purge` also removes the state folder the server used and the AI models. The library is never touched.

## The Linux system service

`sudo sh Ninaivu-<version>-linux-arm64.sh` installs under `/opt/ninaivu` instead, with a system service that runs as its own user, `ninaivu`, not as root: a fault in reading a photograph or a video is then that user's, not the whole machine's. It may still use port 80, and the service has systemd's hardening (no new privileges, `/usr`, `/boot` and `/etc` read-only, a private `/tmp`). Its state is in `/var/lib/ninaivu`, and the default photographs folder is `/var/lib/ninaivu/photos`. A folder elsewhere has to be readable and writable by that user; the installer says so when it is not, for example `sudo setfacl -R -m u:ninaivu:rwX -m d:u:ninaivu:rwX /srv/photos`. Run the maintenance commands as that user too, so the files they write stay its own: `sudo -u ninaivu ninaivu reset-password`.

`sudo systemctl stop ninaivu` lets Ninaivu finish what is in hand: the service allows 45 seconds (`TimeoutStopSec`), and the log names anything that had not stopped by then.

An install from 1.0 ran the service as root, with its data in `/root/.ninaivu`. An upgrade leaves it so, and says so. To move it to its own user: stop it (`sudo systemctl stop ninaivu`), move the data (`sudo mv /root/.ninaivu/* /var/lib/ninaivu/`), give the user the data and the library (`sudo chown -R ninaivu: /var/lib/ninaivu`, and access to the library as above), delete `/etc/systemd/system/ninaivu.service`, and run the installer again with `--photos` and the library folder.

## Docker

For a NAS or a Linux box:

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
cp installers/docker/.env.example installers/docker/.env # set MEDIA_DIR to your
photo folder
docker compose -f installers/docker/docker-compose.yml up -d
```

The family app answers on port 5000 to the whole network; the console on 3000, published on the NAS itself only (`127.0.0.1:3000`). To make the administrator from another computer, open the console through an SSH tunnel or on the NAS, and give the **setup code** printed in `docker logs ninaivu`. The `.env` file is only for folders and ports: everything about how Ninaivu behaves is chosen in the console, and nothing in `.env` undoes it at the next restart. The image leaves out search by description unless built with `--build-arg WITH_AI=1`. Compose builds the image from the checkout; each release also publishes one, `ghcr.io/javajaga-usa/ninaivu:<version>` (and `:latest`), to use in `docker-compose.yml` instead of building. `docker stop` and `docker compose down` give it 45 seconds to finish what is in hand (`stop_grace_period`), rather than Docker's ten. The operations guide covers systemd, reverse proxies, HTTPS and large libraries.

## From a checkout

Running from the source needs Python 3.12 or newer (the installers bring their own).

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
start.cmd ~/Pictures           # Windows: or double-click start.cmd
sh launcher/start.sh ~/Pictures # macOS and Linux
```

The launcher makes a virtual environment, installs what is missing (nothing large: search by description is added later, if you want it), serves HTTPS on port 443 — or 8443 on Linux, where an ordinary user may not use 443, the same port every time — finds free ports and opens your browser. `python -m ninaivu <folder>` runs the server directly, on plain HTTP and port 80 (8080 without the right to use 80), unless given `--https`.

On Windows, `start.cmd` needs no Python beforehand: when it finds no Python 3.12 or newer, it installs Python 3.12 for the current user first (with winget, or from python.org when winget is missing), then carries on. No administrator rights are needed. **Start - Ninaivu Control Panel.vbs** opens the Control Panel once `start.cmd` has run.

On a Mac, double-click **Setup Ninaivu.command** in the Ninaivu folder once instead of using a terminal. It clears the download quarantine, installs Python 3.12 if the Mac has no Python 3.12 or newer (after asking: with Homebrew when the Mac has it, otherwise the installer from python.org, which is run only when it carries the Python Software Foundation's signature, and for which macOS asks your password), installs what Ninaivu needs, and puts two apps in Applications: **Ninaivu**, which starts the library in a Terminal window and asks once which folder holds your photos, and the **Ninaivu Control Panel**, which starts and stops it without one. `sh launcher/start.sh` offers the same Python install. More on the Control Panel.

A checkout keeps its log in `.ninaivu-control/server.log` beside it.

**Videos for guests and share links need ffmpeg.** A phone writes the place a video was filmed into the file, and guests, share links and family members whose copies leave out the home location are given a copy without it, made by ffmpeg. Without ffmpeg on the computer Ninaivu runs on, those people are told the video cannot be shown rather than sent the original. The Docker image includes it; elsewhere, install ffmpeg from the system's packages (on Linux) or Homebrew (on a Mac).

## Running it from the command line

`python -m ninaivu <folder>` (from a checkout) starts the server, and these are the flags worth knowing:

| Flag                                                              |                                                                                 |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <code>--admin USER:PASSWORD</code>                                | make the administrator without a browser                                        |
| <code>--port</code> , <code>--admin-port</code>                   | the two ports (80 — 443 with HTTPS — and 3000 by default)                       |
| <code>--local-only</code>                                         | this computer only; phones cannot reach it                                      |
| <code>--https</code> , <code>--cert</code> , <code>--key</code>   | HTTPS with Ninaivu's own certificate, or yours                                  |
| <code>--ai auto clip light off</code>                             | the image model; <i>auto</i> follows the hardware tier                          |
| <code>--faces</code> , <code>--ocr</code> , <code>--places</code> | turn the passes on from the start                                               |
| <code>--no-watch</code>                                           | do not watch the folders for changes                                            |
| <code>--open-browsing</code> / <code>--private</code>             | whether visitors may look without signing in                                    |
| <code>--lock-roots</code>                                         | confine the folder picker to the library folders                                |
| <code>--admin-host 0.0.0.0</code>                                 | open the console to the network from the start                                  |
| <code>--strict-port</code>                                        | refuse to start rather than move to another port                                |
| <code>--name</code> , <code>--no-mdns</code>                      | the name on the home network ( <code>ninaivu.local</code> ), or none            |
| <code>--map-tiles</code>                                          | OpenStreetMap tiles on the map (they show roughly where photographs were taken) |

When the port is taken by another program, Ninaivu moves to the next free one and says so; `--port` chooses another, `--strict-port` refuses to move.

And the maintenance commands, which work in an installed copy too (the installers put a `ninaivu` command on the computer):

| Command                                                                                  |                                                                                                                                       |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>ninaivu backup --out &lt;folder&gt;</code>                                         | a copy of the index, settings and certificate                                                                                         |
| <code>ninaivu list-backups &lt;folder&gt;</code>                                         | the copies in a folder                                                                                                                |
| <code>ninaivu restore &lt;file&gt;</code>                                                | put one back (Ninaivu stopped)                                                                                                        |
| <code>ninaivu reroot --from &lt;old&gt; --to &lt;new&gt;</code>                          | tell the index the library moved                                                                                                      |
| <code>ninaivu reset-password &lt;name&gt;</code>                                         | a new password for someone, from this computer (a forgotten administrator password)                                                   |
| <code>ninaivu import-lite &lt;file&gt;</code>                                            | bring a Ninaivu Lite household across: people, who sees what, favourites, albums, share links (Ninaivu stopped, after its first scan) |
| <code>ninaivu offsite-restore --recovery &lt;file&gt; &lt;copy&gt; &lt;output&gt;</code> | put the photographs back from an off-site copy, on any computer                                                                       |

The storage check reads the whole library and takes hours on a large one. Until the console has a button for it, `POST /api/admin/scrubber/stop` (signed in to the console) stops it after the file it is reading; a restart does not start it again, and Start carries on from where it stopped.

## Reaching Ninaivu from outside

Ninaivu decides for every request whether it came from home or from the internet. Home is the house's own network (IPv6 included), this computer, Tailscale and WireGuard. Any public address counts as the internet, and so do a forwarded router port, a tunnel or proxy Ninaivu was not told about (forwarded headers from a peer not in `trusted_proxies` and `NINAIVU_TRUSTED_PROXY_ADDRESSES`), and Tailscale Funnel. From the internet:

- profiles set to *tap to enter* do not open, and nobody can browse without signing in;
- plain HTTP straight from a public address is refused, on every path;
- the console refuses every request, whatever `--admin-host` or *Open this console from other devices* say, unless `console_from_internet` (**Advanced settings** → **Remote access**, off by default) is turned on for a tunnel or proxy on purpose;
- 20 wrong administrator passwords in half an hour pause sign-in by username for 30 minutes, then an hour, then two; the computer Ninaivu runs on can still sign in.

Everywhere, uploads are refused while the server's disk would be left with under 2 GB free (phone backups already were), an upload that is really a playlist (`#EXTM3U`, an `ffconcat` list) is refused, and signing out over HTTPS empties the browser's cache of thumbnails.

What the household should set up, once:

1. Reach Ninaivu from away through **Tailscale** (or WireGuard). Forward no port on the router, switch UPnP off there, and keep inbound IPv6 blocked (most routers' default). Never turn on Tailscale Funnel for Ninaivu.
2. Turn on **HTTPS certificates** in Tailscale's admin page, so Ninaivu serves its `.ts.net` name with a certificate phones accept (step by step).
3. Turn on the computer's **firewall**, allowing Ninaivu, and leave *Open this console from other devices* off unless it is needed.
4. A **long administrator password** (14 characters or more, or four random words), kept in a password manager and used nowhere else.
5. A **PIN on every profile**, and open browsing off if anyone not fully trusted uses the Wi-Fi.
6. **Disk encryption** on the computer (FileVault, BitLocker or LUKS): Ninaivu does not encrypt the library or the index on this computer; only the cloud and off-site copies are encrypted.
7. **Two-step sign-in** on the accounts around Ninaivu: Tailscale, Google Drive, the off-site storage service, GitHub.
8. **Keep things up to date**: the operating system, Tailscale, ffmpeg, and Ninaivu when a release comes out (it never updates itself).
9. On a fork of the repository, GitHub's **Dependabot security updates**.

Ninaivu itself has no two-step sign-in yet, and HTTPS is not on by default on the home network; reach it by its Tailscale name at home too, or start it with `--https`, if anyone untrusted may be on the Wi-Fi.

## Tuning to the machine

At every start Ninaivu measures the computer (its cores, memory, graphics processor, whether it is a Raspberry Pi or another single-board computer, and whether the library is on a spinning disk) and picks a profile: **Small box** (a Pi, under 6 GB of memory, or two cores), **Everyday computer**, or **Powerful computer** (eight cores and 16 GB, or a graphics processor and 16 GB). The profile sizes the indexing workers, the analysis threads, the web threads, the image model's batch, the moments described per video, the backup uploads at once and the index cache. **Peak performance** is only ever chosen by hand (or by the desktop panel's Performance mode): up to 95% of the processor and the memory, the rest left for the operating system.

The console's **System → Tuning** page shows what was measured, the profile, what the numbers are expected to use, and every number with where it came from. Any of them can be set outright and put back to automatic; most apply at once, the indexing workers at the next scan, and the web threads and the index cache after a restart, which the page offers. A number given at start (`--workers`, `NINAIVU_SERVER_THREADS`, `NINAIVU_COMPUTE_THREADS`, or written into `config.json` by hand) is kept unless it is set on the page. The choices are saved in `config.json` as `tuning_profile` and `tuning`.

## Updating to a new version

An update replaces the program and nothing else. The photographs are never read or written by an installer, and neither is the state folder (the index, `config.json`, the thumbnails, `setup-code.txt`, the keys) or the AI models.

| How it was installed   | How to update                                                                                              | What is replaced                                                                                                       | What stays as it was                                                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows installer      | run the newer <code>.exe</code>                                                                            | <code>Python</code> , <code>pkgs</code> and <code>bin</code> in the install folder, after Ninaivu is stopped           | the state folder <code>%USERPROFILE%\.ninaivu</code> , the models in <code>%LOCALAPPDATA%\Ninaivu\ai-models</code> , <code>.ninaivu-control</code> , the library |
| macOS app              | drag the new <b>Ninaivu</b> over the old one in Applications (or <code>brew upgrade -cask ninaivu</code> ) | <code>Ninaivu.app</code>                                                                                               | <code>~/ .ninaivu</code> , <code>~/Library/Application Support/Ninaivu</code> , the library                                                                      |
| Linux and Raspberry Pi | run the newer <code>.sh</code> the same way                                                                | the private Python, the commands, the menu entry, the service file (with the same photographs and state folders in it) | the state folder, <code>ai-models</code> , the library                                                                                                           |
| Docker                 | <code>docker compose pull</code> (or <code>build</code> ), then <code>up -d</code>                         | the image                                                                                                              | the <code>ninaivu_state</code> volume and <code>MEDIA_DIR</code>                                                                                                 |
| A checkout             | <code>git pull</code> , then start it again                                                                | the source                                                                                                             | the state folder, <code>.ai-models</code> , the library                                                                                                          |

In Docker, never add `-v` to `docker compose down`: that deletes the state volume, which is the index. The library itself is still untouched.

**The first start of a new version** is the only time anything of the old setup changes: it adds what it needs to `index.db` and `archive.db`, in place. Before that, it copies the whole state folder into `<state>/backups/before-update` (the same verified bundle the scheduled backup makes, the three most recent kept) and notes the version in `<state>/version.json`. If the copy cannot be made (a full disk) Ninaivu does not start, and says so; `NINAIVU_SKIP_UPDATE_BACKUP=1` starts it once without the copy, for someone who has a backup of their own.

**Going back to an older version.** An older Ninaivu refuses to start on an index a newer one has changed in a way it does not know, and names the copy to restore; it never writes over it. To go back: install the older version, then `ninaivu list-backups "<state>/backups/before-update"` and `ninaivu restore <bundle>` (with Ninaivu stopped), which puts back the state from just before the update. Anything done after the update (new albums, names) is not in that copy; the

photographs are not affected either way. Versions up to 1.0.4 do not have this check, so going back to one of those needs the restore first.

## Environment variables and the state folder

Some settings also read an environment variable ( `NINAIVU_STATE_DIR` , `NINAIVU_AI_ENGINE` , `NINAIVU_HARDWARE_TIER` , `NINAIVU_FACES` , ...; the full list is in `ninaivu/server/config.py` ). Folders and ports from the environment always apply; anything about behaviour only fills in what nobody chose in the console. A number out of range, from the environment or typed into `config.json` by hand, is brought within its limits (one from the environment is named in the log); a value of the wrong kind in `config.json` is set aside with a message and Ninaivu starts anyway. The state directory ( `~/.ninaivu` , or `$XDG_DATA_HOME/ninaivu` , or `NINAIVU_STATE_DIR` ) holds the index, the thumbnails and `config.json` . The AI models live in `.ai-models` in a checkout; an installed copy keeps them in a folder of the user's own, which an upgrade does not replace ( `%LOCALAPPDATA%\Ninaivu\ai-models` on Windows, `~/Library/Application Support/Ninaivu/ai-models` on a Mac, `~/.local/state/ninaivu/ai-models` on Linux). `NINAIVU_AI_MODELS_DIR` or the `ai_models_dir` setting moves them; the Docker image sets the former to `/data/state/ai-models` , on the state volume.

## Further reading

- Operations and production — Docker, systemd, reverse proxies, large libraries, backups.
- The tray, the Control Panel and HTTPS
- Backup and recovery and encryption and the recovery file
- Moving to another machine
- Contributing and security